

Answers learning task 19

2026 – 2027

Exercise 11.1: An article in *Concrete Research* [“Near Surface Characteristics of Concrete: Intrinsic Permeability” (1989, Vol. 41)] presented data on compressive strength x and intrinsic permeability y of various concrete mixes and cures. Summary quantities are $n = 14$, $\sum y_i = 572$, $\sum y_i^2 = 23,530$, $\sum x_i = 43$, $\sum x_i^2 = 157.42$, and $\sum x_i y_i = 1697.80$. Assume that the two variables are related according to the simple linear regression model.

- Calculate the least squares estimates of the slope and intercept. Estimate σ^2 . Graph the regression line.
- Use the equation of the fitted line to predict what permeability would be observed when the compressive strength is $x = 4.5$.
- Give a point estimate of the mean permeability when compressive strength is $x = 3.3$.
- Suppose that the observed value of permeability at $x = 3.3$ is $y = 45.6$. Calculate the value of the corresponding residual.

Answer

- (a) $\hat{y} = 48.0130 - 2.3298x$, $\hat{\sigma}^2 = 1.8436$ (b) 37.5289
 (c) 40.3246 (c) 5.2754

Exercise 11.4: An article in *Technometrics* by S. C. Narula and J. F. Wellington [“Prediction, Linear Regression, and a Minimum Sum of Relative Errors” (Vol. 19, 1977)] presents data on the selling price and annual taxes for 24 houses. The data are in the following table.

Sale Price / 1000	Taxes (local, school, county) /1000	Sale Price / 1000	Taxes (local, school, county) /1000
25.9	4.9176	30.0	5.0500
29.5	5.0208	36.9	8.2464
27.9	4.5429	41.9	6.6969
25.9	4.5573	40.5	7.7841
29.9	5.0597	43.9	9.0384
29.9	3.8910	37.5	5.9894
30.9	5.8980	37.9	7.5422
28.9	5.6039	44.5	8.7951
35.9	5.8282	37.9	6.0831
31.5	5.3003	38.9	8.3607
31.0	6.2712	36.9	8.1400
30.9	5.9592	45.8	9.1416

- Assuming that a simple linear regression model is appropriate, obtain the least squares

fit relating selling price to taxes paid. What is the estimate of σ^2 ?

- (b) Find the mean selling price given that the taxes paid are $x = 7.30$.
- (c) Calculate the fitted value of y corresponding to $x = 5.6039$. Find the corresponding residual.
- (d) Calculate the fitted $\hat{y}_i$ for each value of x_i used to fit the model. Then construct a graph of $\hat{y}_i$ versus the corresponding observed value y_i and comment on what this plot would look like if the relationship between y and x was a deterministic (no random error) straight line. Does the plot actually obtained indicate that taxes paid is an effective regressor variable in predicting selling price?

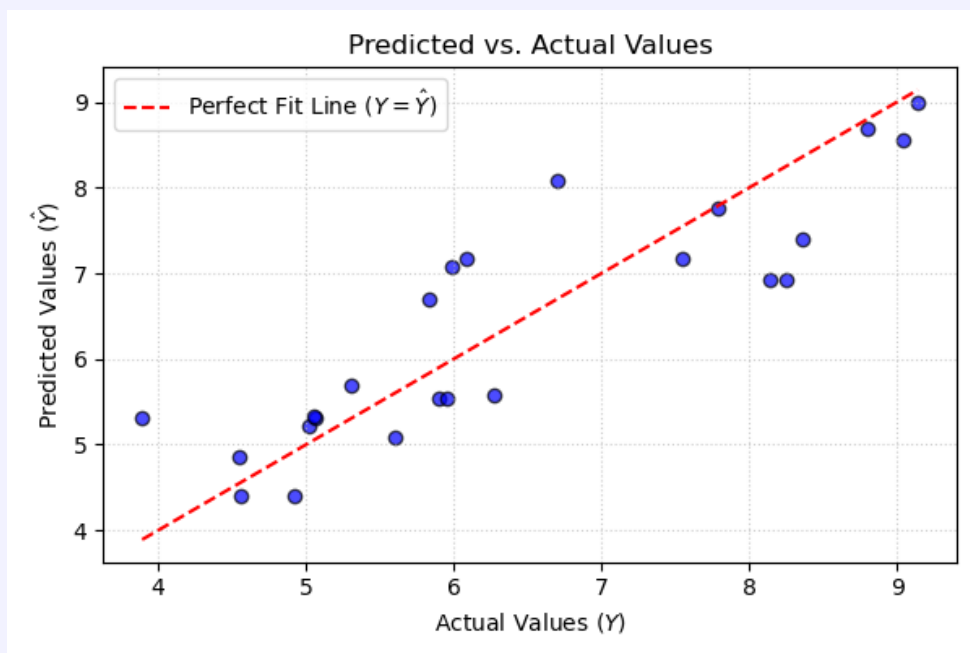
Answer

(a) $\hat{y} = 13.3202 + 3.3244x$, $\hat{\sigma}^2 = 8.7678$

(b) 37.5881

(c) 31.9496, -3.0496

- (d) Plot of $\hat{y}_i$ vs y_i : notice the data points are close to the ideal line, so taxes paid is an effective regressor variable in predicting the sale price.



Exercise 11.6: The following table presents the highway gasoline mileage performance and engine displacement for DaimlerChrysler vehicles for model year 2005 (U.S. Environmental Protection Agency).

Carline	Engine Displacement (inch ³)	MPG (highway)
300C/SRT-8	215	30.8
CARAVAN 2WD	201	32.5
CROSSFIRE ROADSTER	196	35.4
DAKOTA PICKUP 2WD	226	28.1
DAKOTA PICKUP 4WD	226	24.4
DURANGO 2WD	348	24.1
GRAND CHEROKEE 2WD	226	28.5
GRAND CHEROKEE 4WD	348	24.2
LIBERTY/CHEROKEE 2WD	148	32.8
LIBERTY/CHEROKEE 4WD	226	28
NEON/SRT-4/SX 2.0	122	41.3
PACIFICA 2WD	215	30.0
PACIFICA 4WD	215	28.2
PT CRUISER	148	34.1
RAM 1500 PICKUP 4WD	500	18.7
RAM 1500 PICKUP 4WD	348	20.3
SEBRING 4-DR	165	35.1
STRATUS 4-DR	148	37.9
TOWN & COUNTRY 2WD	148	33.8
VIPER CONVERTIBLE	500	25.9
WRANGLER/TJ 4WD	148	26.4

- Fit a simple linear model relating highway miles per gallon (y) to engine displacement (x) in cubic inches using least squares.
- Find an estimate of the mean highway gasoline mileage performance for a car with 175 cubic inches engine displacement.
- Obtain the fitted value of y and the corresponding residual for a car, the Neon, with an engine displacement of 134 cubic inches.

Answer

Note: The Neon has an engine displacement of 122 in³.

(a) $\hat{y} = 39.1555 - 0.0402x$

(b) 32.1176

(c) 34.2491, 7.0509

Exercise 11.8: An article in the *Journal of Sound and Vibration* (Vol. 151, 1991, pp. 383.394) described a study investigating the relationship between noise exposure and hypertension. The following data are representative of those reported in the article

y	1	0	1	2	5	1	4	6	2	3
x	60	63	65	70	70	70	80	90	80	80

y	5	4	6	8	4	5	7	9	7	6
x	85	89	90	90	90	90	94	100	100	100

Answer

Note: Replace $(y, x) = (6, 90)$ in the top row with $(y, x) = (6, 80)$.

(a) $\hat{y} = -9.8131 + 0.1715x, \hat{\sigma}^2 = 1.9822$

(b) 5.6204

Exercise 11.18: Show that in a simple linear regression model the point $(\bar{x}, \bar{y})$ lies exactly on the least squares regression line.

Answer

$$\hat{\beta}_0 + \hat{\beta}_1 \bar{x} = (\bar{y} - \hat{\beta}_1 \bar{x}) + \hat{\beta}_1 \bar{x} = \bar{y}$$